COMP34812 COURSEWORK · EVIDENCE DETECTION

Evidence Detection With a BiLSTM Baseline and a Transfer-Initialised DeBERTa Ensemble

We model claim-evidence relevance as a binary sequence-pair classification problem. The final coursework submission uses one non-transformer system (`Solution B`) and one transformer system (`Solution C`), both evaluated on the official closed-track train/dev data and locked through explicit run records.

TASK SUMMARY

Given a claim and a candidate evidence snippet, predict whether the evidence is relevant.

- Track: `Evidence Detection (ED)`
- Setting: closed-track coursework data only
- Submission pair: one `B` method and one `C` method

DATASET

Official Train/Dev Data

Split	Rows	Negative	Positive
Train	21,508	15,654	5,854
Dev	5,926	4,286	1,640
Combined	27,434	19,940	7,494

The positive class is about **27.3%** of the combined train+dev data. Claims are very short, while evidence snippets are materially longer, which makes thresholding and false-positive control important.

SEQUENCE CHARACTERISTICS

Pair Lengths

Statistic	Claims	Evidence
Mean words (train)	5.60	31.13
Mean words (dev)	5.62	31.20
Median words	5	28

SUBMISSION LOCK

Canonical Systems

Submission paths are locked in `experiments/decisions/D011_submission_lock.md`.

- **Solution B**: ESIM-style BiLSTM, threshold `0.60`
- **Solution C**: transfer-initialised DeBERTa-v3 3-seed ensemble, threshold `0.50`

SOLUTION B

ESIM-Style BiLSTM

Goal: provide a strong non-transformer baseline that still models claim-evidence interaction rather than classifying each sequence independently.

1. **Embedding**
   Lowercase, lightweight cleanup, FastText 300d vectors via gensim.
2. **Shared BiLSTM Encoding**
   Claim and evidence are encoded with the same sequence encoder.
3. **Co-Attention**
   Cross-attention aligns both sequences and exposes token-level interaction.
4. **Composition + Pooling**
   Enhanced representations pass through a second BiLSTM and mean/max pooling.
5. **Classification**
   MLP head outputs the binary relevance score.

ACCURACY	BINARY F1
0.8137	**0.6824**

MACRO F1	MCC
0.7753	**0.5529**

The locked `B` path is threshold-sensitive. The default threshold `0.50` underperformed the tuned operating threshold `0.60`, indicating that calibration mattered materially for minority-class behaviour.

SOLUTION C

Transfer-Initialised DeBERTa-v3 Ensemble

Goal: maximise transformer performance while keeping the path stable, reproducible, and clearly measurable.

1. **Base Model**
   Start from `MoritzLaurer/DeBERTa-v3-base-mnli-fever-anli`.
2. **Pair Encoding**
   Cross-encoder on claim/evidence pairs with max length `256`.
3. **Stable Training**
   Float32 path, `lr=1e-5`, `epochs=3`, `weight_decay=0.01`, warmup `0.06`.
4. **Ensemble**
   Train seeds `42`, `52`, `62` and average probabilities.
5. **Locked Path**
   Use default threshold `0.50` for the submission system.

ACCURACY	BINARY F1
0.8890	**0.8133**

MACRO F1	MCC
0.8671	**0.7384**

Transfer initialisation produced a much larger gain than continuing broad tuning on the plain baseline. A three-seed ensemble improved robustness, while pushing to five seeds gave weak returns.

DEV RESULTS

Against Official Baselines

System	Acc	Bin F1	Macro F1	MCC
Official ED LSTM baseline	0.8058	-	0.7083	0.4674
Official ED BERT baseline	0.8745	-	0.8348	0.6748
Locked Solution B	0.8137	0.6824	0.7753	0.5529
Locked Solution C	0.8890	0.8133	0.8671	0.7384

LIMITS AND RISKS

What This Work Does Not Solve

- The dataset is imbalanced, so headline accuracy can hide minority-class failures.
- Neither system provides explanatory rationales for why evidence was judged relevant.
- False positives can surface irrelevant evidence; false negatives can hide useful evidence.
- The closed-track setting prevents use of external retrieval or evidence-ranking data.

ERROR ANALYSIS

Observed Failure Modes

- **Solution B:** over-predicts the positive class at the default threshold, which hurts both macro F1 and MCC.
- **Solution C:** still makes `451` false positives and `207` false negatives on dev at the locked threshold.
- Many remaining errors appear to come from evidence that is semantically related to the claim but not actually strong enough to count as relevant evidence.

WHY C LEADS

Locked Submission Comparison

Metric	Solution B	Solution C
Accuracy	0.8137	0.8890
Binary F1	0.6824	0.8133
Macro F1	0.7753	0.8671
MCC	0.5529	0.7384

`Solution C` is the primary submission system because it is ahead on every tracked dev metric. `Solution B` remains valuable because it satisfies the second required approach category and provides the non-transformer comparison system.

DEMO PATHS

Marker-Facing Reproducibility

- **Solution B notebook:** `solution_b_demo_inference.ipynb`
- **Solution C notebook:** `solution_c_5_seed_ensemble_development.ipynb`
- **Submission lock:** `experiments/decisions/D011_submission_lock.md`
- **Test-day runbook:** `ops/TEST_DAY_RUNBOOK.md`

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